



Semester 1 Examinations 2018

Exam Code(s) 1SD1, 1MDM1, 1IF1, 1MF1, 1MCR1, 1SD3
Exam(s) Higher Diploma in Software Design and Development
Higher Diploma in Software Design and Development,
Industry Stream
Masters in Information Technology
Masters in Software Design and Development
Masters in Digital Media
Masters Clinical Research

Module Code(s) CT511
Module(s) Database Development

Paper No.
Repeat Paper

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Instructions: Answer **questions 1 and 2** and 2 other questions
All questions carry equal marks

Duration 2 Hours
No. of Pages 10
Discipline(s) Information Technology
Course Co-ordinator(s)

Requirements:

MCQ
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper

Other Materials

Release to Library: Yes ☒ No ☐

PTO

QUESTION 1

a) Create an **EERD** for the following:

Dooley Meats (identified by company id, contact and location) comprises several distinct subsidiaries: Dooley Chicken, Dooley Fish and Dooley Beef. The parent company Dooley Meats plans in the future to add additional subsidiaries, finances permitting. Each subsidiary has an id, COO and location.

Each subsidiary operates several departments (HR, Finance, Manufacturing and Sales) which, given a lack of funding, sometimes overlap in their operations. More departments are planned within the next 5 years.

Each department is run by only one administrator. Some administrators (identified by PPS and name) run one department. Administrators (some of whom run no departments) are further categorised on the basis of their contract type; hourly, part time or full time. These are the only possible contract categories. Administrators can also be categorised based on their grades; Grade1, Grade 2 and Grade 3. There are plans to add further grades later. Some Grade 3 administrators supervise at least one Grade 2 administrator. While the Grade 1 administrator takes a number of courses to upskill; all Grade 2 administrators are supervised by one Grade 3. Courses have a unique ID, a name and a duration and need at least one person registered, for them to run.

All administrators are assigned to a supervisor, who is identified by his supervisor id and name. The Hourly administrator has a specific charge/hour, whilst the Full Time administrator records the date hired and the Part Time administrator records the day.

A department can be shared by multiple subsidiaries. Each department is identified by a department id and contact.

Dooley Chicken records chicken numbers, Dooley Fish records fish stock and Dooley Beef records tag number.

(17 marks)

b) Identify whether or not the following are in **2NF** and/or **3NF**. If they are in **2NF**, explain why. If they are not in **3NF**, explain and decompose them to 3NF.

EMPLOYEE (EMPLOYEE-No, SECTION-No)
CLASS (COURSE-No, SECTION-No, ROOM)
CLASS (COURSE-No, ROOM, CAPACITY, COURSE-NAME)

(6 marks)

c) What is meant by the term **Data Abstraction**?

(2 marks)

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QUESTION 2

a). Write the SQL to create a table called **test**, having the following columns; ID is the primary key and is an integer, Name can only hold a maximum of 25 characters, Salary contains salaries ranging from 90,000.55 to 120000.99 and DOB is a date. All columns cannot contain null values.

(2 marks)

b). Add a new column called County to the test table. It should take 2 characters only. It should default to SO.

(1 mark)

Note : The remaining SQL queries relate to the tables; **STAFF** and **BANK**.

STAFF

staffno	fname	lname	position	sex	dob	salary	county	code
1	Mary	Walsh	Teller	F	1962-01-01	35000.00	Galway	93-53-79
2	Tom	Connor	Manager	M	1960-10-11	120000.88	Dublin	93-53-62
3	John	Wilde	Teller	M	1967-10-12	62000.88	Sligo	93-53-68
4	Ann	Casey	HR	F	1970-09-20	36000.95	Galway	93-53-79
5	Pearse	Curry	Teller	M	1972-11-04	38000.95	Galway	93-53-79
6	Joe	Murphy	Manager	M	1961-10-22	102000.98	Sligo	93-53-68
7	David	Ford	Teller	M	1970-10-11	42000.98	Dublin	93-53-62

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BANK

Sort	Name	Location
93-53-79	AIB	Main Street, Galway
93-53-78	AIB	Quay Street, Galway
93-53-62	BOI	OConnell Street, Dublin
93-53-63	BOI	Parnell Street, Dublin
93-53-64	BOI	Pearse Street, Dublin
93-53-65	BOI	Tara Street, Dublin
93-53-66	BOI	Edgeworth Street, Dublin
93-53-67	BOI	Flaming Street, Dublin
93-53-68	BOI	Main Street Sligo

c). Write the SQL which calculates the number of staff per county, where the average salary from each county is greater than the average salary of all staff (regardless of county). The following is displayed:

County average salary Vs Average of all salaries
2 staff in the Dublin offices, where the average Dublin salary of 81000.8 is greater than the average of all salaries
1 staff in the Galway offices, where the average Galway salary of 900000.00 is greater than the average of all salaries
2 staff in the Sligo offices, where the average Sligo salary of 820000.92 is greater than the average of all salaries

(3 marks)

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d). Write the SQL which displays the following information of those staff whose surname contains a C OR those staff whose salary is less than the average of all salaries (note: only the first 3 characters of the surname are shown. Also, watch the way it is ordered).

Abbreviated surname	county
Con***	Dublin
For***	Dublin
Wal***	Galway
Cas***	Galway
Cur***	Galway
Wil***	Sligo

(3 marks)

e). Select all records where the surname is NOT an item in the returned list where the staff number (e.g. 1, 2, 3) is equal to the count of the number of staff in each county (note: this involves multiple subqueries).

(3 marks)

f). Write the SQL which displays the following:

String manipulation
Mary has 4 characters in their name. Reversed, their name is yraM
Tom has 3 characters in their name. Reversed, their name is moT
John has 4 characters in their name. Reversed, their name is nhoJ
Ann has 3 characters in their name. Reversed, their name is nnA
Pearse has 6 characters in their name. Reversed, their name is esraeP
Joe has 3 characters in their name. Reversed, their name is eoJ
David has 5 characters in their name. Reversed, their name is divaD

(2 marks)

PTO

g). Write the SQL which displays the names of those staff having the top salaries in each county.

Top salaries per county

Tom Connor has the top of 120000.88 in county Dublin

Pearse Curry has the top of 38000.95 in county Galway

Joe Murphy has the top of 102000.98 in county Sligo

(3 marks)

h). Write the SQL where the highest paid staff in each of their respective positions is presented as follows (note: this is a join) :

fname	lname	salary	position	name	location
Ann	Casey	36000.95	HR	AIB	Main Street, Galway
Tom	Connor	120000.88	Manager	BOI	OConnell Street, Dublin
John	Wilde	62000.88	Teller	NULL	NULL

(3 marks)

i). Write the SQL which revises the salary of those staff whose salary is less than ALL of the minimum salaries for each position. In the event of this happening, the revised salary is 2.5 times the current salary amount, otherwise it remains the same.

Display the following:

fname	lname	position	salary	Revised
Mary	Walsh	Teller	35000.00	The salary is changed from 35000.00 to 87500.00
Tom	Connor	Manager	120000.88	The salary remains the same at 120000.88
John	Wilde	Teller	62000.88	The salary remains the same at 62000.88
Ann	Casey	HR	36000.95	The salary remains the same at 36000.95
Pearse	Curry	Teller	38000.95	The salary remains the same at 38000.95
Joe	Murphy	Manager	102000.98	The salary remains the same at 102000.98
David	Ford	Teller	42000.88	The salary remains the same at 42000.88

(3 marks)

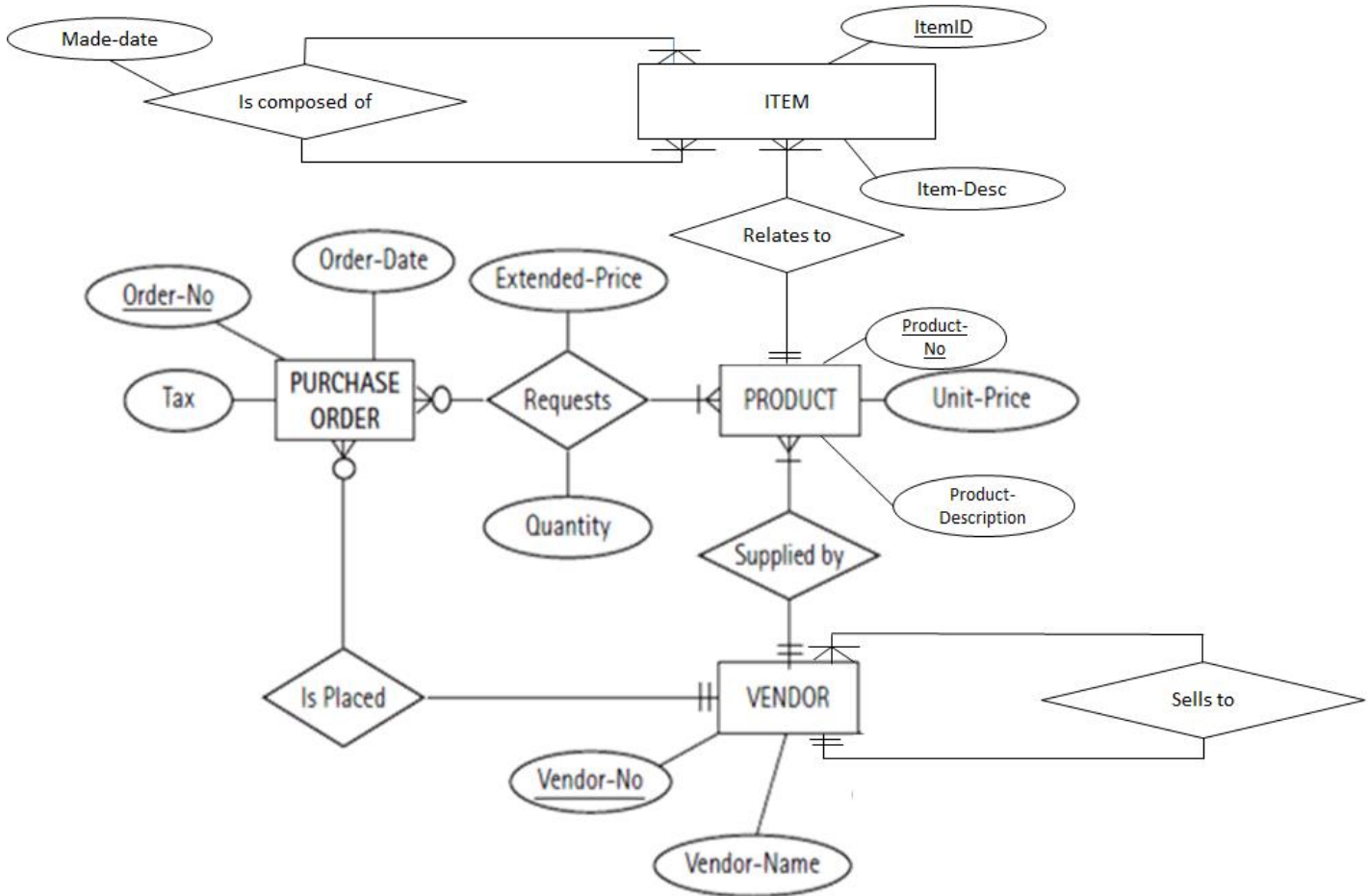
j). Write the SQL to add a new row to the staff table. Include the following data: Mick Forrest, with a Staff ID of 8, is a manager who was born on the 30th October 1982. His current salary is 90000 and he will be working at the Galway branch (93-53-79).

(2 marks)

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QUESTION 3

a) Map the following **ERD** to relations using the **short hand approach**.



(11 marks)

b) Compare and contrast **left join**, **inner join** and **right join**. Use sample tables and sample SQL code to highlight the differences.

(9 marks)

c) Why perform **Denormalisation**?

(3 marks)

d) Explain what is meant by **non-loss decomposition**.

(2 marks)

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QUESTION 4

a) Bring the following table to **2NF** explaining why it is not in 2NF (use the **5 step approach**).

CourseID	CourseName	Date-Completed	Instructor	ClassID	Room	Capactiy
C101	Java	23/5/2018	Dr. Pam Shares	CL1	R101	54
C101	Java	21/7/2018	Dr. Ethan Hawke	CL2	R102	34
C102	Maths	22/3/2019	Dr. Liam Skelton	CL1	R101	54
C103	PHP	21/8/2019	Dr. Dr. Ethan Hawke	CL2	R103	67

(15 marks)

b) A sequence of database operations that satisfies the **ACID** properties is perceived as a single logical operation on the data. Explain what is meant by the 4 parts of ACID (use examples).

(8 marks)

c) Explain the importance of minimizing **Data Redundancy** ?

(2 marks)

QUESTION 5

a) Explain why the following table is in **2NF**. Bring it to **3NF** explaining why it is not in 3NF.

EmpID	EmpName	EmpEirCode	EmpAddress	EmpCity	BankAc	Amount
E101	Tom Aleet	109987H	9 The Oaks, Salthill	Galway	1567	1230.45
E102	John McCarthy	263879J	10 The Elm, Gort	Galway	2345	3423.32
E103	Liam Skills	231516L	21 The Fir, Loughrea	Galway	5678	4323.29

(10 marks)

b) Draw and then discuss the '**Three Schema Architecture**', explaining the layers and the different data independences therein.

(8 marks)

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c) Bring the following table to **BCNF (Boyce-Codd Normal Form)** using the updated information : “An employee works in one department, yet a department has many employees working for it.”

Draw the **Functional Dependency Diagram (FDD)** and **map to relations** using the **graphical approach**.

<u>ProductID</u>	<u>Dept</u>	<u>EmpID</u>	<u>Nationality</u>
P145	Sales	101	Irish
P145	Engineering	102	German
P234	Design	103	American
P688	Sales	105	Irish
P234	Engineering	102	German
P123	Sales	104	Austrian

(7 marks)

QUESTION 6

a) What is meant by **Database Table Partitioning**?

Perform **Vertical** and **Horizontal Partitioning** on the following table.

Note: Whilst the first two products are the most popular and regularly referenced, reports are frequently required for the product and its respective review.

Key	Product Name	Short Description	Review	Picture
01	Americano @ Starbucks	Black, no sugar	I'd buy again	
02	BB @ Seattle's Best	Black, no sugar	The best	
03	TB @ Zoka Coffee	Black, no sugar	It's okay	
04	BC @ Coffee	Black, no sugar	Never again	

(8 marks)

b) Compare and contrast the 4 **Data Integrity Constraints**. Use examples to show your understanding.

(8 marks)

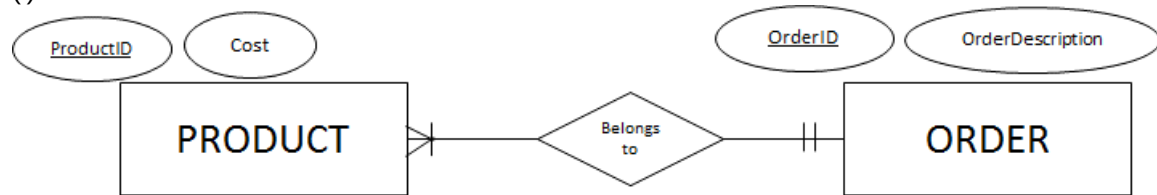
c) Discuss 3 strategies for **Data Fragmentation** in **Distributed Database Systems**.

(5 marks)

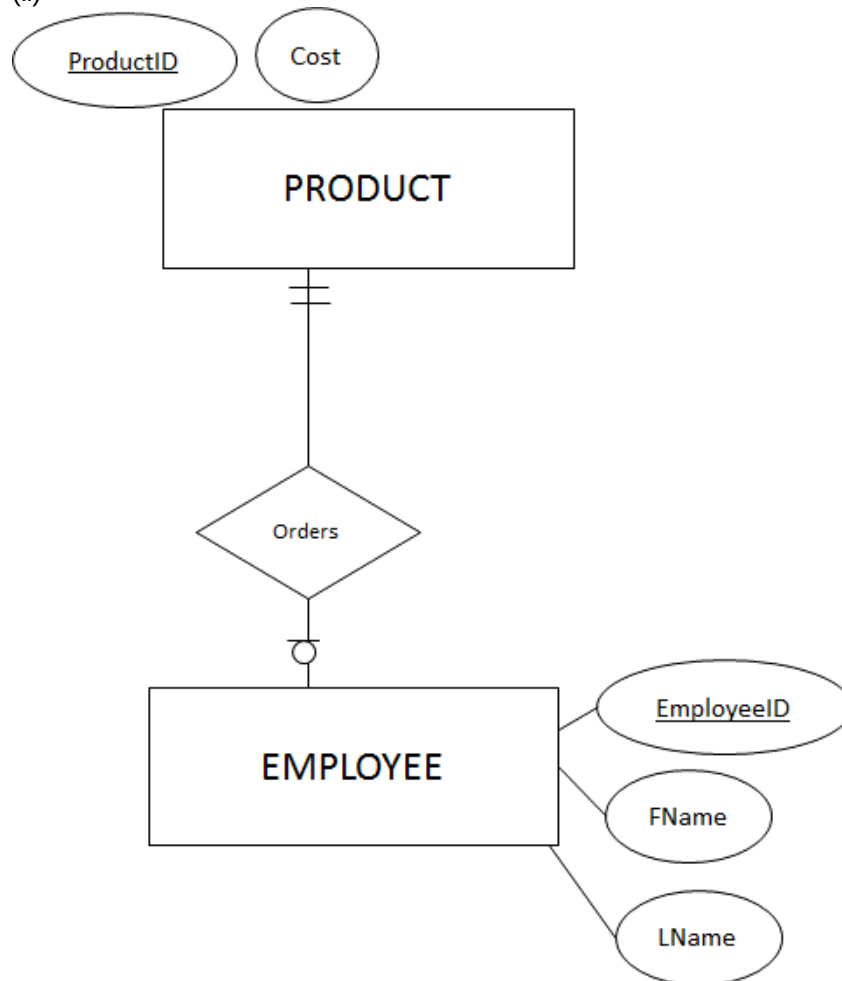
PTO

c) Map the following **ERDs** to relations using the short hand approach and then **denormalise**.

(i)



(ii)



(4 marks)